MesoNet: Optimizing Spiking Neural Networks with Dynamic Saddle Distributions

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Abstract

Artificial neural network (ANN) clustering algorithms require hundreds of watts, considerably limiting their abilities across numerous energy-critical applications. A developing technology, spiking neural networks (SNNs) offer a promising alternative, requiring mere milliwatts by mimicking biological neurons on neuromorphic chips. However, the prevalent spike-timing-dependent-plasticity (STDP) learning rule struggles to scale for practical tasks, reducing accuracy by up to 40% compared to ANNs. This paper addresses these challenges with a two-step approach: mathematically investigating SNN learning behavior with dynamical systems theory, before developing an algorithm for improved generalization. We identify an optimal learning regime at the edge of chaos and introduce two algorithms- variable plasticity and triangulated attribution- maintaining these optimal learning conditions. Notably, these mechanisms dynamical form and annihilate saddle points to enhance representations. Implemented in a split-and-merge architecture, our network MesoNet implements these algorithms to achieve performative improvements in accuracy and information retention. This includes achieving a 56.6% accuracy on CIFAR-10, 18.57% points above the previous SOTA SNN. MesoNet lies within 5.05% points of state-of-the-art ANNs while consuming half (47.23%) the energy. This architecture demonstrates significant potential for transforming computation in applications including deep-space exploration, medical implants, and remote sensing.

Code Availability: https://github.com/blizzard-labs/NMCtests

1 Introduction

Ás AI applications permeate into various parts of technology, unsupervised learning is gaining increased awareness in the edge computing use-case. The need for online learning is expanding into real-world applications such as medical implants, robotics, space exploration, and the internet of things.[23] However, modern unsupervised algorithms based on artificial neural networks (ANNs) are computationally hungry and incur massive energy costs [17] becoming a real obstacle for when energy usage is limited.

Concurrently, spiking neural networks (SNNs), regarded as the third generation of neural networks [13], are being designed to explore the intricacies of intelligence and brain function by replicating

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the behavior of biological neurons[20] through incorporation of biologically-inspired features. These networks are built upon two key elements: spiking neurons and synapses, which are organized into hierarchical layers to form cohesive networks.

Recently, SNNs have gained substantial attention in the research community due to their unique advantages, such as low power consumption [20], event-driven processing, online learning, and massive parallelism [17]. These abilities stem from their sparse spike-based computation and communication, which can be exploited to achieve higher computational efficiency by specialized neuromorphic hardware [7]. SNNs have potential to solve many problems regarding unsupervised edge computing due to these characteristics. However, as a nascent field of study, SNNs have yet to achieve accuracy comparable to state-of-the-art (SOTA) ANNs.[12]

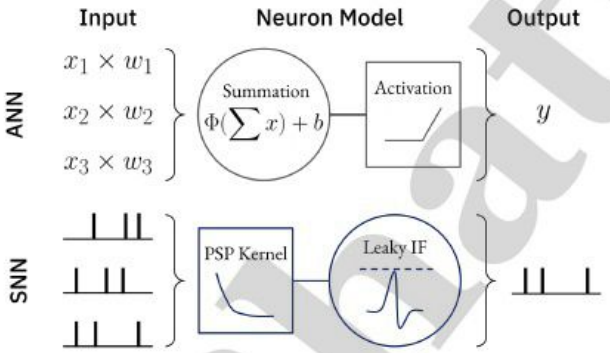


Figure 1: SNNs process information asynchronously through sparse biological spikes with spike timings encoding information.

1.1 Overview of SNN Architectures

Currently SNNs can be broadly divided into three major categories: supervised, unsupervised, and converted. Supervised SNNs denote the network's use of labelled datasets to guide training, specifically through following the gradient of their loss function. Various strategies have been developed to do this including backpropagation through time and use of surrogate gradients[15] as the spiking nature of neurons makes them non-differentiable; however none have yet met the performance of classical ANNs. Unsupervised SNNs denote the algorithmic training of SNNs without the use of labelled datasets. Algorithms such as spike-timing dependent plasticity (STDP) and DECOLLE [10] adjust weights based on biological synaptic plasticity to learn the intrinsic structures of input samples. Converted SNNs utilize pretrained ANNs and transfer weights [5] to SNNs to circumnavigate training difficulties. This paper will focus on the unsupervised class of SNNs for their implications in online learning and edge computing.

1.2 Spike Timing Dependent Plasticity

Most modern unsupervised SNNs are trained through spike-timing dependent plasticity due to its computational resourcefulness, proven effectiveness, and bio-plausibility [2]. Intuitively based upon the common adage, "neurons that fire together wire together," STDP-based learning rules modify synaptic weights interconnecting a pair of pre and post-synaptic neurons based on the degree of correlation between the respective spike timings. This process of STDP builds on biological concepts of Long Term Potentiation (LTP) and Long Term Depression (LTD) [21], and can be described as the following:

$$w = \sum_{\text{pre}} \sum_{\text{post}} W(t_{\text{post}} - t_{\text{pre}}) \tag{1}$$

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$$W(t) = \begin{cases} A_{\text{pre}} \times e^{-\frac{\Delta t}{\tau}} & \Delta t > 0 \\ A_{\text{post}} \times e^{\frac{\Delta t}{\tau}} & \Delta t < 0 \\ 0 & \text{else} \end{cases} \tag{2}$$

where t_{pre} and t_{post} are the spike timings of the pre and post-synaptic neurons respectively, A_{pre} and A_{post} are the learning rates governing plasticity, and τ is a learning time constant. LTP can be defined by $W(\Delta t)$ when Δt is positive and LTD when Δt is of a negative value. However, the locally-based learning in individual synapses means that STDP has various limitations: In a preliminary study by Diehl and Cook (2015) [3], standard conductance-based STDP was utilized to train a fully connected shallow network to a sub-optimal performance level of 95% accuracy on the toy dataset MNIST with an extremely slow learning speed of 900,000 iterations to complete training.

Additionally, catastrophic forgetting [16], a phenomenon observed in continual learning and STDP where knowledge of old tasks are lost when new ones are learned is frequently observed within STDP based neural networks. As such, the performance of STDP-trained networks seems to be limited beyond basic tasks like handwritten digit recognition. To address some of these issues, convolutional SNNs, similar to their deep learning counterparts, were proposed to learn hierarchical input representations [11] and improve scalability. However, it remains to have lower performance than SOTA ANNs on most popular benchmark datasets. As described by Rathi et al. (2022) [19], STDP trained SNNs face two primary issues, that if surpassed, "would enable a new generation of low-cost (area/power/resource requirement), local and unsupervised learning framework, as opposed to their non-spiking counterparts (ANNs)." These two problems include:

1. The scalability of SNNs remains to be unknown without the typical issues of information loss.
2. It is not yet known if STDP alone can enable deeper layers to learn complex input representations and generalize on low-level features and clusters.

1.3 Contributions

Here, we design a STDP-based SNN that improves in all learning efficiency (speed), network sensitivity and stability, and high-level generalizations to create a spatiotemporal neural architecture for energy efficient unsupervised learning. To summarize, this work proposes three main contributions:

Uncovered nonlinear learning dynamics in SNN systems. Investigated information loss and chaos in deep layers. Developed a novel plasticity based learning algorithm and mathematically proved its improved capability

1. Provide a theoretical framework describing a system of STDP synapses as a bifurcation dynamical system of learning rate, extending the work of Zhang et al. (2021)
2. Investigate information loss and nonlinear dynamics in deep layers.
3. Develop a novel plasticity-based learning architecture MesoNet and demonstrate its performance in generalization on complex recognition tasks going past MNIST.

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2 Theoretical Findings

2.1 Preliminaries

2.1.1 The LIF Neuron Model

In order to begin deconstructing the LIF-STDP SNN network, we must first consider the most basic case, the 1:1 connection, which can be defined as the connection between two LIF neurons by a STDP synapse. We review a general form of the LIF model as follows, provided by (Zhang, 2021) [24]:

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -v(t) + Rf(I(t)) \tag{3}$$

where $I(t) = (I_1(t), ...I_M(t))^\intercal$ denotes the M-dimensional input signal, $u(t)$ indicates the membrane potential of the concerned neuron at time t, τ_m and R are positive-valued hyper-parameters with respect to membrane time and membrane resistance, respectively, and f represents the aggregation function, usually expressed in linear formation, $f(I(t)) = \sum_{j=1}^{M} W_j I_j(t)$, in which W_j denotes the learnable connection weight corresponding to the j-th input channel. Based on the *spike response model* scheme [6], the LIF equation has a general solution with the boundary condition $v_{rest} = 0$ as follows:

$$v(t) = \sum_{j=1}^{M} W_j [\int_{t'}^{t} \exp(\frac{t'-s}{\tau_m}) I_j(s)\mathrm{d}s] \tag{4}$$

where t' denotes the last firing time $t' = \max\{s|v(s) = v_{firing}, s < t\}$. An output stimulus $S(t)$ is generated whenever the membrane potential is reached. $v(t)$ reaches a certain threshold v_{firing} (firing threshold). We formulate this procedure using a spike excitation function

$$f_e : u \to S, \text{where} S(t) \triangleq \lfloor \frac{v(t)}{v_{firing}} \rfloor \tag{5}$$

After firing, the membrane potential is instantaneously reset to a lower value u_{rest} (rest voltage). Some researchers take the delayed firing behaviors in neuroscience, *i.e.*, add an absolute refractory period Δ_{abs} [8] or a refractory kernel $\epsilon_{ref}(\tau_m)$ [4]. Thus, Eq. (2) can become

$$v(t) = \sum_{j=1}^{M} W_j [\int_{t'}^{t} \exp(\frac{t'-s}{\tau_m} I_j(s)\mathrm{d}s] + \epsilon_{ref}(\tau_m) * S(t) \tag{6}$$

where $*$ denotes the convolution operation and $\epsilon_{ref}(\tau_m) * S(t)$ is the refractory response of a neuron.

2.1.2 Neural Encoding

Compared to ANNs, SNNs use spikes for representation of data, and as such, real input data should be converted into a spiking version through neural encoding prior to entering the network. There are two popularized forms of neural encoding in current literature: *rate-based and temporal*. In temporal encoding, data is encoding through the temporal distance between firing events [7]. In rate based coding, information is represented through the mean firing rate of a neuron over a given time period. In poisson rate coding, a subcategory of the latter, the normalized analogue value is compared to a randomly generated number from a poisson distribution. The neuron fires if the value is greater than the number and otherwise remains inactive [18]. Although temporal encoding is considered more energy efficient for their "data-rich" sparse spiking, it faces issues with noise amplification for the

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reduced number of spikes. Rate-based encoding faces few of such issues and can also be practically recorded by a Dynamic Vision Sensor [1], making it the optimal algorithm for the purposes of this project.

2.1.3 Spike-timing dependent plasticity

The prevalent algorithm, Spike-Timing-Dependent Plasticity (STDP) is a biologically inspired learning rule that modifies the strength of synaptic connections based on the relative timing of pre-synaptic and post-synaptic spikes.

Let t_{pre} and t_{post} denote the firing times of a pre-synaptic and a post-synaptic neuron, respectively. The change in synaptic weight Δw is defined as a function of the time difference $\Delta t = t_{post} - t_{pre}$ such that:

$$\Delta w = \begin{cases} A_+ \cdot \exp(\frac{-\Delta t}{\tau_l}) & \text{if} \Delta t > 0 \\ A_- \cdot \exp(\frac{\Delta t}{\tau_l}) & \text{if} \Delta t < 0 \\ 0 & \text{if} \Delta t = 0 \end{cases} \tag{7}$$

Here, $A_+ > 0$ and $A_- < 0$ are the maximum amplitudes of synaptic potentiation and depression, respectively, and τ_l are the time constants governing the decay of the learning rate. This temporally asymmetric rule reflects experimental observations that causally correlated spike pairs typically lead to long-term potentiation (LTP), wheras anti-causal pairs lead to long-term depression (LTD).

2.2 The 1:1 Connection

In order to map a direct relationship between input and output spike times, we consider Poisson rate encoding where k (spikes / τ_m) denotes the input rate to a neuron i and $x = t' - t$ denotes the output spiking rate. With this definition, we can simplify the 1:1 neuron to

$$v(t) = \sum_{j=0}^{M} W_j k x \exp(\frac{-x}{\tau_m}) \tag{8}$$

We can further simplify this to a single-input channeled-neuron given by

$$f(x) = u(t) = Wkxe^{\frac{-x}{\tau_m}} \tag{9}$$

where $f_i(x)$ and $f_j(x)$ denotes the presynaptic and postsynaptic neurons respectively. However, as given by Eq. (5) and Eq. (6), the STDP and LIF models are inherently recursive as spike timing differences lead to further weight updates which can, in turn, influence spike timing differences. To build a deterministic relationship for STDP which can expand to a larger system, we must form a definition of the timing difference Δt for any given time t.

One method of doing so would be directly setting $f(x)$ to v_{thresh} as given by:

$$f(x) = wkxe^{\frac{-x}{\tau_m}} - \theta = 0 \tag{10}$$

where $\theta = v_{thresh}$. Through direct calculation, we find that Δt can be expressed in terms of the Lambert W function (ω) which solves for the inverse of $W(z) = ze^z$.

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$$
\begin{aligned}
x e^{\frac{-x}{\tau_m}} &= \frac{\theta}{wk} \\
\text{setting } y = \frac{-x}{\tau_m}, \;\; y e^y &= \frac{-\theta}{\tau_m wk} \\
\text{thus, } \Delta t(t) = t' - t = x &= -\tau_m \omega(\frac{-\theta}{\tau_m wk})
\end{aligned} \tag{11}
$$

where y is the intermediary Lambert-inverse function. However, this definition for spike timing differences, utilizing the Lambert W-function remains non-ideal for the purposes of this project and the analysis of SNN dynamical systems. A fundamental limitation arises from the multi-valuedness of the function. For a given complex value z, there exists an infinite number of branches $W_k(z)$, $k \in \mathbb{Z}$, each corresponding to a distinct solution to the ewuation $we^w = z$. When applied to the characteristic equation of a standard linear time-delay system in the form

$$\dot{x}(t) = Ax(t) + Bx(t - \tau) \tag{12}$$

the eigenvalues of the system satisfy

$$\lambda_k = \frac{1}{\tau} W_k(\tau B e^{A\tau}), \tag{13}$$

where each branch W_k contributes a distinct dynamic mode. In contrast to ordinary differential equations (ODEs), which yield a finite spectrum of eigenvalues, systems expressed with the Lambert W function formulation require managing an *infinite spectrum of delay-induced modes*. This complicates modal analysis and characterization of the dynamical systems. Additionally, there exists a branch-cut along the interval $(-\infty, -1/e]$ on the real axis (separating the principle branch W_0 and the lower real-valued branch W_{-1}. In the vicinity of this cut, small perturbations in the input z can cause discontinuous jumps between branches, undermining the smoothness assumptions required in our linear stability analysis.

2.3 Quadratic Spike Approximation

In order to overcome the issues associated with utilizing the Lambert W function, we propose a quadratic spike approximation which provides a deterministic method of computing spike timing differences. These results can be extended to network implementations as well as dynamical systems analysis. The truncated approximation for spike timing can be given by

$$f(x) \approx w \cdot k \cdot x \cdot T_N(x) = \theta \tag{14}$$

where $T_N(x) = \sum_{n=0}^{N} \frac{(-x/\tau_m)^n}{n!}$ and n satisfies the condition $(n+1)! \geq 8e\tau_m^2$. Here, n provides the minimum-ordered approximation required to approximate spike timing within the operating interval. You can refer to the full proof in the appendix.

2.4 Local Linear Stability Analysis

We can now extend these findings to develop a complete model for the behavior of the 1:1 connection. Consider the STDP equation from Eq. (5). This can be rearranged and simplified to the basic form

$$\Delta w(\Delta t(w)) = \text{sgn}(\Delta t(w)) \cdot \eta \cdot \exp(-\text{sgn}(\Delta t(w)) \cdot \frac{\Delta t(w)}{\tau_m}) \tag{15}$$

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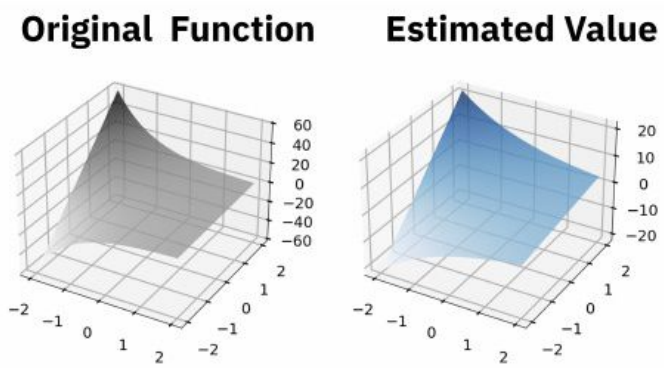


Figure 2: A Visualization of Quadratic Spike Approximation over sample domain.

where sgn is the "sign" function such that $\text{sgn}(x) = |x|/x$. We can also recast STDP dynamics by defining the Poisson-based spike time difference

$$\Delta t(w) = n(\frac{1}{k_{post}(w)} - \frac{1}{k}) \tag{16}$$

so that the sign of Δt now directly reflects whether the post-synaptic inter-spike interval is longer or shorter than the pre-synaptic one. (Here n indexes the spike pair). For a given connection, the fixed point is reached when $\Delta t(w) = 0$. This is because the magnitude of update is nonzero unless the exponential factor acts on a zero argument. Thus, the fixed point weight is provided by

$$w^*(k)^2 \text{exp}(\frac{-k}{\tau_m}) \rightarrow w^* = \frac{\theta e^{k/\tau_m}}{k^2} \tag{17}$$

Going on to analyze the linear stability about the fixed point, we find the final linearization as

$$f'(w^*) = -\frac{\eta}{\tau_m} \frac{n}{w^* k \cdot (1 - \frac{k}{\tau_m})} \tag{18}$$

The evolution of f(w) depends entirely on the factor $1 - \frac{k}{\tau_m}$ and the learning rate parameter η. In many biophysical contexts, the membrane time constant τ_m is small compared to the effective time scale set by k_{pre}, so that typically,

$$\frac{k}{\tau_m} > 1 \Rightarrow 1 - \frac{k}{\tau_m} < 0 \tag{19}$$

This makes $f'(w^*) > 0$, so that in the standard STDP-LIF SNN 1:1 connection, small perturbations in weight grow exponentially and the fixed point w^* is entirely unstable. (Complete calculations available in appendix.)

2.5 Deeper Multilayer Dynamics

Now with an understanding of the governing dynamics underlying the 1:1 connection, we can extend this model into a multilayer context. Consider a simple three-layered network, with one input ($x \in \mathbf{R^{NxN}}$), one hidden ($h \in \mathbf{R^{NxN}}$), and one ouput layer ($y \in \mathbf{R^{NxN}}$). Matrices W_2 and W_1 are the connecting weights between the hidden and the output, and the input and the hidden, respectively. To review, for a given connection- from a presynaptic firing rate (k) to a postsynaptic firing rate (k_{post}), the fixed point condition is reached when,

$$\Delta t = 0 \Rightarrow \frac{1}{k_{post}} = \frac{1}{k} \Rightarrow k_{post} = k \tag{20}$$

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In what follows, we assume the network learns to "align" the frequencies in successive layers, so that at a fixed point, as per the standard STDP function, and we have $k^x = k^h = k^y \equiv k$. Explicitly, let us write that

- W^1 be the A x B weight matrix from layer x to h
- W^2 be the B x C weight matrix from layer h to y

The firing rate of a hidden neuron j is determined by its net input,

$$I_j^1 = \sum_{i=1}^{A} W_{ij}^1 \cdot k_i^x \tag{21}$$

and we employ the bio-physical "activation function" derived from the last section

$$k_j^h = -\tau_m \omega(\frac{-\theta}{I_j^1 \tau_m}) \tag{22}$$

Similarly, for an output neuron k in y, we have

$$I_k^2 = \sum_{j=1}^{B} W_{jk}^2 \cdot k_j^h \text{ and } k_k^y = -\tau_m \omega(-\frac{\theta}{I_j^2 \tau_m}) \tag{23}$$

For each individual connection, the STDP update is given by

$$\Delta W_{ij}^1 = \phi\eta(\frac{1}{k_j^h} - \frac{1}{k_i^x}) \text{ and } \Delta W_{jk}^2 = \phi\eta(\frac{1}{k_y^k} - \frac{1}{k_h^j}) \tag{24}$$

with

$$\phi(z) = \text{sgn}(z)\exp(-\frac{|z|}{\tau_m}) \tag{25}$$

Thus, the overall system dynamics are governed by the collection of differential equations

$$\begin{aligned} \frac{\mathrm{d}W_{ij}^1}{\mathrm{d}t} &= F_{ij}^1(W^1, W^2) \\ \frac{\mathrm{d}W_{jk}^2}{\mathrm{d}t} &= F_{jk}^2(W^1, W^2) \end{aligned} \tag{26}$$

In order to study the stability for the multilayer case, we define a state vector v that collects all weight elements

$$\begin{pmatrix} \text{vec}(W^1) \\ \text{vec}(W^2) \end{pmatrix} \tag{27}$$

where, near the fixed point, we write

$$v = v^* + \delta v \text{ and } \frac{\mathrm{d}}{\mathrm{d}t}\delta v = \mathbf{J}\delta v \tag{28}$$

where the Jacobian matrix, represented by $\mathbf{J}$ is the block matrix

$$\mathbf{J} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \tag{29}$$

Here, J_{11} is the partial derivative of the F^1 updates with respect to W^1, and J_{12} comes from any (indirect) dependence of F^1 on W^2. Note that in our hypothetical feed forward architecture, the

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hidden layer's firing rates depend only on W^1, so we expect J_{12} to vanish. J_{21} accounts for the fact that F^2 updates depend on the hidden-layer firing rates k^h, which in turn, depend on W^1. J_{22} is the partial derivative of F^2 with respect to W^2. Recall, once again, that for the single layer case,

$$f'(w^*) = \frac{-\eta n}{\tau_m w^* k(1 - \frac{k}{\tau_m})} \tag{30}$$

In the multidimensional case, the linearization can be lifted through the following Jacobian components

$$J_{11} \approx \alpha_1 I_{N1}, \;\; J_{12} \approx \alpha_2 I_{N2} \tag{31}$$

where I_{N1} and I_{N2} are the identity matrices of appropriate dimensions (the number of connections in layer x and layer h, respectively) and the local gradients are given by

$$\alpha_1 = \frac{\eta n}{\tau_m W^{1*} k(\frac{k}{\tau_m} - 1)} \quad \alpha_2 = \frac{\eta n}{\tau_m W^{2*} k(\frac{k}{\tau_m} - 1)} \tag{32}$$

Note the negative component in the single-connection expression can by disregarded by the sign change when $1 - \frac{k}{\tau_m} < 0$. The off-diagonal block J_{21} now arises because the updates F^2 depend on the hidden firing rates of k^h (and hence W^1); its exact form is given by

$$J_{21} = \frac{\partial F^2}{\partial k^h} \frac{\partial k^h}{\partial W^1} \tag{33}$$

In our considered feedforward network, the block J_{12} is assumed to be 0, as discussed earlier. Thus the full Jacobian is block-lower-triangular.

$$\mathbf{J} = \begin{pmatrix} J_{11} & 0 \\ J_{21} & J_{22} \end{pmatrix} \tag{34}$$

Using the identity from the Schur complement,

$$\det \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \det(B_{22} - B_{21} B_{11}^{-1} B_{12}). \tag{35}$$

we know that the eigenvalue λ is a number such that $Ax = \lambda x$. From which, we get $\det(A - \lambda I_N) = 0$. In our case, B_{12} is a zero matrix and hence, we get,

$$\det(B - \lambda I_N) = \det(\begin{pmatrix} B_{11} & 0 \\ B_{21} & B_{22} \end{pmatrix} - \lambda I_N) = \det \begin{pmatrix} B_{11} - \lambda I_k & 0 \\ B_{21} & B_{22} - \lambda_{N-k} \end{pmatrix} \tag{36}$$

Therefore, $\det(B - \lambda I_N) = \det(B_{11} - \lambda I_k) \times \det(B_{22} - \lambda I_{N-k})$. Through this reasoning, we can conclude that the eigenvalues of the full multilayer system's Jacobian are given by the union of the eigenvalues of J_{11} and J_{22}. And in thus, in our simplified (and symmetric) case, we have:

- For every $x \to h$ weight, every connection contributes an eigenvalue

$$\lambda_1 = \alpha_1 = \frac{\eta n}{\tau_m w^{1*} \cdot k(\frac{k}{\tau_m} - 1)} \tag{37}$$

- For every $h \to y$ weight, every connection contributes an eigenvalue

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$$\lambda_2 = \alpha_2 = \frac{\eta n}{\tau_m w^{2*} \cdot k(\frac{k}{\tau_m} - 1)} \tag{38}$$

We can rewrite this as

$$\lambda_1 = \lambda_2 = \frac{\eta n k^2 e^{-k/\tau_m}}{\tau_m \theta(\frac{k}{\tau_m} - 1)} = \frac{\eta \cdot nke^{-k/\tau_m}}{\tau_m \theta(\frac{k}{\tau_m} - 1)} \tag{39}$$

Moreover, if we denote by N_1 the number of connections from $x \to h$ and by N_2 the number of connections from $h \to y$, the complete spectrum of the Jacobian would be given by

$$\text{spec}(\mathbf{J}) = \{\lambda_1, ..., \lambda_1, \lambda_2, ..., \lambda_2\} \tag{40}$$

2.5.1 Bifurcational Classification

With this analysis, we can reinterpret the network's weight dynamics as a bifurcational problem in which the learning rate, η, plays the role of the bifucational parameter. Recall that in our derivation for the multilayer system. the local dynamics near each connection can be written as

$$\frac{\mathrm{d}\partial w}{\mathrm{d}t} = f'(w^*)\delta w \tag{41}$$

with

$$f'(w^*) = \alpha\eta \tag{42}$$

Because the eigenvalues of the Jacobian are exactly these slopes, the linearized term in $\lambda = \alpha\eta$. Thus,

$$\frac{\mathrm{d}\delta w}{\mathrm{d}t} = \lambda\delta w = \alpha\eta\delta w \tag{43}$$

Notice that with this system:

- For $\eta = 0$, the eigenvalue $= 0$. In this degenerate case, the fixed point is neutrally stable.
- For $\eta > 0$, (the standard case) we have $\lambda > 0$ provided that the constant α is positive. Hence, small perturbations from the fixed point grow exponentially.
- For $\eta < 0$ the sign of λ reverses, so that the fixed point becomes locally stable.

In this sense, the learning rate η controls the stability of the fixed point. The bifurcation point occurs at $\eta = 0$, as a codimension-1 stability switch. This concludes the proof for bifurcational classification (Theorem 2 as listed on the poster).

2.6 Defining an Optimal Learning Regime

Given this deeper understanding of the dynamical and bifurcational systems underlying SNN learning, we propose an optimal operating region for the network within the (η, w) parameter space, utilizing an analysis of chaos and network-wide Lyapunov exponents. Within the context of the previously defined multi-layered system, a natural candidated Lyapunov energy function would be one that could "measure" the mismatch between post- and pre-synaptic rates. One can define an energy function as the squared error in this timing mismatch:

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$$V(w) = \frac{1}{2}[\Delta t(w)]^2 \tag{44}$$

Notice that at the fixed point (where $k_{post}(w) = k$), we have $\Delta t(w) = 0$ and hence $V(w) = 0$. In the multilayer network, where the entire weight matrix is denoted by $\mathbf{w}$, we extend this to

$$V(\mathbf{w}) = \frac{1}{2}\sum_i [\Delta t_i(w)]^2 \tag{45}$$

where each $\Delta t_i(\mathbf{w})$ is defined for the i-th connection. This function first minimizing energy at the "desired" state, similar to the potential energy of physical systems and provides a scalar measure of "error" that we can use to assess the system's behavior as a function of its learning rate η and the current weight values w. Additionally, we can imagine augmenting the energy with a term that penalizes "too large" weight changes, similar to the kinetic energy of physical systems. Now, an effect energy can be defined as

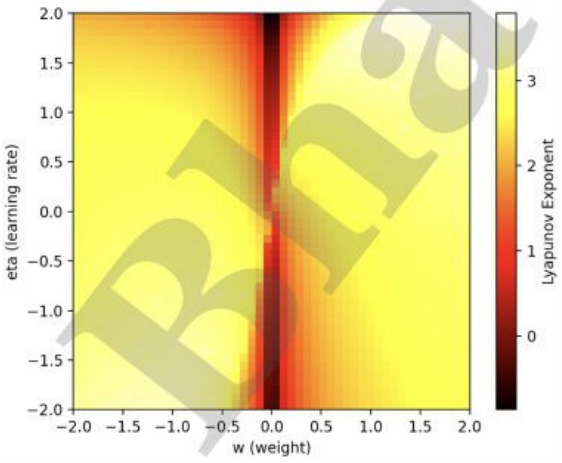


Figure 3: A Lyapunov exponent heatmap of a multilayer STDP network, quantifying divergence and demonstrating the "edge of chaos"

$$E(w, \eta) = \frac{1}{2}[\Delta t(w)]^2 + \frac{\kappa}{2}(f'(w))^2 \tag{46}$$

where $f'(w)$ is the local gradient of the update function and, from our previous analysis, we found

$$f'(w) \approx \alpha\eta \;\text{ with }\; \alpha = \frac{n}{\tau_m w^* k(\frac{k}{\tau_m} - 1)} \tag{47}$$

In this picture, the second term penalizes regions where the instantaneous growth rate is too high, thereby "selecting" an optimal region where learning is neither too slow (if η is very small) nor too unstable (if η is too large). This function also delineates the boundary between a chaotic phase, where signal propogation expands and folds space in a chaotic manner, and an ordered phase. This would be where the learning function contracts space in an ordered manner and back-propagated gradients exponentially vanish. We will continue to consider this Lyapunov-like energy function in future discussion regarding the MesoNet algorithms. However, for now, we can go on to define the dominant Lyaponov exponent, in its fully expanded form, as

$$l_{max}(w, \eta) = \alpha\eta = -\frac{\eta}{\tau_m^2 w(1 + \omega(-\frac{\theta}{wk\tau_m}))\omega(-\frac{\theta}{wk\tau_m})} \exp\left(-\frac{\left|\frac{1}{-\tau_m\omega(-\frac{\theta}{wk\tau_m})} - \frac{1}{k}\right|}{\tau_m}\right)\text{sgn}\left(\frac{1}{-\tau_m\omega(-\frac{\theta}{wk\tau_m})} - \frac{1}{k}\right) \tag{48}$$

By analytically mapping $E(w, \eta)$ and l_{max} over the (w, η) plane, we identify an optimal acting region as that in which the energy is minimized and the largest Lyapunov exponent is near zero (or slightly negative), indicating a balance between rapid learning ad stability. In our system, ensuring that the dominant Lyapunov exponent is not too positive (or is slightly negative) guarantees that weight updates remain controlled, thereby defining an optimal regime of operation in terms of both learning rate η and the weight w.

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3 Optimizing SNN Learning

To briefly review, from our analysis of SNN dynamical systems, we have unlocked a few novel insights into the underlying functionality behind learning. We have observed that

- A STDP multi-layered network is a fully dissapative system where minute perturbations can disrupt the system such that it diverges over time
- The learning parameter η acts as a codimension-1 bifurcation parameter within the network and can effectively determine whether the system is dissapative, conservative, or diffusive in regards to synaptic weights
- There exists an optimal learning regime in the (w, η) space where Lyapunov exponents are near zero such that the network can effectively switch between chaotic and stable phases.

Additionally, recall the standard STDP learning rule common amongst under unsupervised SNN applications. This learning rule can be reformulated through a trace implementation in neuromorphic applications:

Algorithm 1 STDP Learning Rule (Trace Implementation) across Multilayered Network

Input: Initial weight matrix $W \in \mathbb{R}^{N_{\text{pre}} \times N_{\text{post}}}$ (small random values), learning rate η, time constant τ_m, threshold θ, input firing rate k_{in}, and presynaptic rate k_{pre}.
for each trial $t = 1, \ldots, T$ **do**
  for each synapse (i, j) **do**
    for $t = 0$ to T with step Δt **do**
      $x \leftarrow x \cdot e^{-\Delta t/\tau_x}$
      $y \leftarrow y \cdot e^{-\Delta t/\tau_y}$
      if $t \in T_{\text{pre}}$ **then** ▷ Pre-synaptic spike at time t
        $x \leftarrow x + 1$
        $w \leftarrow w + A_+ \cdot y$ ▷ Potentiation
      end if
      if $t \in T_{\text{post}}$ **then** ▷ Post-synaptic spike at time t
        $y \leftarrow y + 1$
        $w \leftarrow w - A_- \cdot x$ ▷ Depression
      end if
      $w \leftarrow \text{clip}(w, w_{\min}, w_{\max})$
    end for
  end for
end for

With required parameters: Pre-synaptic spike times T_{pre}, Post-synaptic spike times T_{post} Synaptic weight w, Learning rates A_+, A_- Time constants τ_x, τ_y for pre/post traces Initial traces $x \leftarrow 0$, $y \leftarrow 0$ Time step Δt, Total simulation time T. Through modifying synaptic trace variables upon every post-synaptic and pre-synaptic firing, this implementation discretizes the original update function

$$\Delta w = \begin{cases} A_+ \cdot \exp(\frac{-\Delta t}{\tau_l}) & \text{if} \Delta t > 0 \\ A_- \cdot \exp(\frac{\Delta t}{\tau_l}) & \text{if} \Delta t < 0 \\ 0 & \text{if} \Delta t = 0 \end{cases} \tag{49}$$

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However, as noted by various other authors, spike-timing-dependent-plasticity is an entirely local learning process and can lead to the network not being capable of scaling to larger layers. These findings have been supported by our previous analysis of the multilayer system, which suggests that a multilayer system is fully dissapative and leads to weight divergence overtime into various local minima- all determined through biological rules of long term potentiation (LTP) and long term depression (LTD). In order to optimize SNN learning, we propose two algorithms that can dynamically form and annhilate saddles within the (w, η) space such that the network remains within the previously defined optimal learning regime. In the following we will review their explicit functionality.

3.1 Variable Plasticity

The first algorithm we propose, *Variable Plasticity*, enables the network to dynamically switch between chaotic and stable modes. Rather than keeping η fixed, we allow it to evolve according to dynamics that tend to pull towards a minimum value η_{min} while also being modulated by the local weight change $w(\Delta t)$. We define the plasticity evolution as

$$\tau_d \frac{\mathrm{d}\eta}{\mathrm{d}t} = -(\eta - \eta_{\min}) + \frac{1}{\eta_{\max} - w(\Delta t)} \tag{50}$$

where τ_d is a decay (or adaptation) constant, η_{min} and η_{max} set lower and upper bounds on η, and $w(\Delta t)$ denotes the standard local weight change computed from the timing error. This equation makes the effective learning rate "variable" so that when the local weight update is large (or the weight is close to its upper bound), the effective η is reduced, promoting stability. Algorithmically, variable plasticity can be defined as the following

Algorithm 2 Variable Plasticity

1: **Input:** Initial weight matrix $W \in \mathbb{R}^{N_{\text{pre}} \times N_{\text{post}}}$, initial learning rate matrix $\eta \in \mathbb{R}^{N_{\text{pre}} \times N_{\text{post}}}$ (small random values), parameters $\eta_{\min}$, $\eta_{\max}$, τ_m, decay constant τ_d, threshold θ, input firing rate k_{in}, and presynaptic rate k_{pre}, triangulation factor ϵ, and similarity constant κ.
2: **for** each trial $t = 1, \ldots, T$ **do**
3: Present the input spike pattern.
4: Compute presynaptic spike times $\{t_i^{\text{pre}}\}$ and postsynaptic spike times $\{t_j^{\text{post}}\}$.
5: **for** each synapse (i, j) **do**
6: Update the synaptic weight based upon STDP:

$$w_{ij} \leftarrow w_{ij} + \Delta w_{ij}^{\text{std}}.$$

7: Update the learning rate (variable plasticity) using a discrete approximation of

$$\tau_d \frac{d\eta_{ij}}{dt} = -\Big(\eta_{ij} - \eta_{\min}\Big) + \frac{1}{\eta_{\max} - w_{ij}(\Delta t_{ij})};$$

e.g.,

$$\eta_{ij} \leftarrow \eta_{ij} + \frac{\Delta t}{\tau_d}\left[-\Big(\eta_{ij} - \eta_{\min}\Big) + \frac{1}{\eta_{\max} - w_{ij}(\Delta t_{ij})}\right].$$

8: **end for**
9: **end for**

With required parameters: Pre-synaptic spike times T_{pre}, Post-synaptic spike times T_{post}

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Synaptic weight w, Learning rates A_+, A_- Time constants τ_x, τ_y for pre/post traces Initial traces $x \leftarrow 0$, $y \leftarrow 0$ Time step Δt, Total simulation time T.

3.2 Triangulated Attribution

The second algorithm we propose, *Triangulated Attribution*, enables synapses to attribute local spike timing differences to external synapses. When two synapses of a layer share the same Δt, rather than updating itself independently, we "triangulate" an attribution onto a candidate synapse in the preceeding layer that is most relevant to both. Concretely, suppose synapses i and j in a layer satisfy the condition, $\Delta t_i \approx \Delta t_j$, where the maximum tolerable difference can be set as a parameter. Define a function R for each candidate synapse p in the pre-synaptic neuron(s) of the local synapse.

1. **Relevance Score**: For each candidate p and for the pair (i, j), define

$$R_p^{i,j} = \eta_p \exp(-\frac{|w_{pi} - w_{pj}|}{\kappa}) \tag{51}$$

   where η_p is the current plasticity at the candidate synapse, w_{pi} and w_{pj} are the weights from neuron p to the post-synaptic neurons corresponding to synapses i and j, and κ is a scaling constant that controls the sensitivity to the weighted distance between target and source synapses.

2. **Selection and Update**: Identify the candidate synapse p^* such that $p^* = \arg\max_{p \in \mathbf{P}}(R_p^{i,j})$. Then, we perform an additional update on the candidate synapse p^*

$$\Delta w_{p*} = \epsilon w(\Delta t) \tag{52}$$

Altogether, Triangulated Attribution is algorithmically defined by the pseudocode in Alg. (3). Notice that to identify sets of synapses with related spiking, quadratic spiking approximation allows for exact estimation within $O(n\log(n))$ time through use of Newton's Method. Additionally, we utilize Dijkstra's algorithm for identifying candidate synapses through a relevance graph.

With required parameters: Pre-synaptic spike times T_{pre}, Post-synaptic spike times T_{post} Synaptic weight w, Learning rates A_+, A_- Time constants τ_x, τ_y for pre/post traces Initial traces $x \leftarrow 0$, $y \leftarrow 0$ Time step Δt, Total simulation time T.

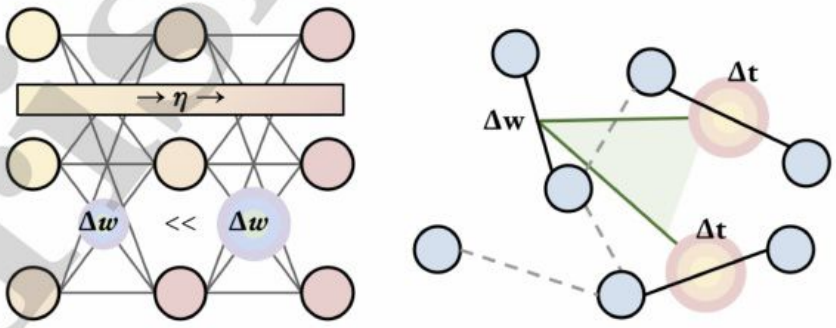


Figure 4: Altogether, Variable Plasticity and Triangulated Attribution enable MesoNet to enhance learning through attribution and encapsulate learned uncertainties with plasticity.

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Algorithm 3 Triangulated Attribution via Dijkstra's Algorithm

Require:

$S \leftarrow \{s_1, s_2, \ldots, s_n\}$ ▷ Set of synapses in layer L with identical Δt

For each $s \in S$, let Δw_s be its computed weight update and let η_s be its plasticity

$G = (V, E)$ ▷ Directed graph representing connectivity from layer $L-1$ to layer $L-2$

For each edge $e \in E$, let $d(e)$ be its physical (or functional) distance and let η_e be its associated plasticity factor

γ ▷ Scaling constant for triangulated update

Ensure: An additional update Δw_{target} applied to a target synapse s^* in the layer preceding the pre-synaptic neurons of S

Initialization:

for each synapse $s \in S$ **do**

    Identify pre-synaptic neuron p_s in layer $L-1$

end for

Let $P \leftarrow \{p_s : s \in S\}$ ▷ Set of unique pre-synaptic neurons

Build Relevance Graph:

The graph $G = (V, E)$ represents connections from layer $L-2$ to layer $L-1$.

For each edge $e \in E$ connecting neuron $u \in V$ to neuron $v \in P$, define the *cost* (or inverse relevance) as:

$$C(e) = \frac{d(e)}{\eta_e}$$

▷ A low cost corresponds to high relevance.

Dijkstra Search for Highest Relevance Path:

for each pre-synaptic neuron $p \in P$ **do**

    Run Dijkstra's algorithm on G (with edge cost $C(e)$) starting from p

    Let $C_p(x)$ denote the minimal cumulative cost from p to candidate neuron $x \in V$

end for

Aggregate Relevance Scores:

For each candidate neuron $x \in V$, define the combined cost

$$C_{\text{combined}}(x) = \sum_{p \in P} C_p(x)$$

▷ Alternatively, one might take the maximum or minimum over $p \in P$.

Identify the candidate neuron

$$x^* \leftarrow \arg\min_{x \in V} C_{\text{combined}}(x)$$

▷ This is the upstream neuron with maximum (combined) relevance to all pre-synaptic neurons.

Triangulated Update:

Compute the aggregate weight update from the synapses in S,

$$\Delta w_S = \frac{1}{|S|} \sum_{s \in S} \Delta w_s$$

Set the additional update as

$$\Delta w_{\text{target}} = \gamma \cdot \Delta w_S$$

Apply Δw_{target} to the synapse connecting x^* to the appropriate pre-synaptic neuron (or distribute it among relevant connections)

return Δw_{target}

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3.3 MesoNet Central Conjecture

These modifications introduce a dynamic learning rate that adapts to the local weight change- damping learning when weights approach critical levels- and a cooperative "triangulated" update that leverages similarity in timing error across synapses to propogate a corrective signal to a related upstream connection. Together, they are designed to counteract the instability of the standard STDP rule by effectively narrowing the operating window to one near marginal stability, where learning is efficient yet controlled. In this section, we will provide the reasoning regarding Variable Plasticity's and Triangulated Attribution's functionality.

3.3.1 Reconsidering Dynamical Systems

We begin by making revisions to the previously described dynamical systems under the standard STDP learning rule. In the revised model, the weight dynamics are modified by a variable learning rate η as well as an additional update term that "triangulates" attribution when a similar timing error occurs. In particular, we write the net weight update as

$$\frac{\mathrm{d}w}{\mathrm{d}t} = f(w,\eta) = \eta\phi(\Delta t(w)) + \epsilon g(w, \Delta t(w)) \tag{53}$$

with

$$\phi(\Delta t) = \mathrm{sgn}(\Delta t) \cdot \exp(-\frac{|\Delta t|}{\tau_m}) \tag{54}$$

and where the triangulated term (scalled by a small constant ϵ) is active when two synapses in a layer share the same Δt and effectively "pull" an upstream connection. Based upon this assumption, we can predict that the triangulated term vanishes at the fixed point. (That is, at the fixed point we have $\Delta t(w^*) = 0$ and the extra update is zero.) Simultaneously, the learning rate itself evolves via

$$\tau_m \frac{\mathrm{d}\eta}{\mathrm{d}t} = -(\eta - \eta_{min}) + \frac{1}{\eta_{max} - w} \tag{55}$$

Thus, the fixed point of the full revised system is the pair (w^*, η^*) satisfying the timing error, which remains the same as for standard STDP ($\Delta t(w^*) = 0$ and $k_{post}(w^*) = k$), so that w^* is given by

$$w^* = \frac{\theta e^{k/\tau_m}}{k^2} \tag{56}$$

and the steady state for the learning rate

$$0 = -(\eta^* - \eta_{min}) + \frac{1}{\eta_{max} - w^*} \quad \Rightarrow \quad \eta^* = \eta_{min} + \frac{1}{\eta_{max} - w^*} \tag{57}$$

3.3.2 Linear Stability Analysis

W now linearize the dynamics about the fixed point. Define small perturbations $\delta w = w - w^*$ and $\delta\eta = \eta - \eta^*$. Linearizing the weight update yields

$$\frac{\mathrm{d}\delta w}{\mathrm{d}t} \approx \frac{\partial f}{\partial w}|_{(w^*,\eta^*)} \cdot \delta w + \frac{\partial f}{\partial \eta}|_{(w^*,\eta^*)} \cdot \eta \tag{58}$$

At the fixed point $\Delta t(w^*) = 0$, and -assuming continuity- the derivative of the standard STDP term is, as in our previous analysis,

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$$\frac{\partial}{\partial w}[\eta\phi(\Delta t(w))]_{(w^*,\eta^*)} = \eta^*\phi'(0)\frac{\mathrm{d}\Delta t}{\mathrm{d}w}|_{w^*} \tag{59}$$

with $\phi'(0) = -\frac{1}{\tau_m}$ (since locally the exponential behaves linearly) and where we define $A = \frac{\mathrm{d}\Delta t}{\mathrm{d}w}|_{w^*}$. Thus, the standard term contributes

$$-\frac{\eta^*}{\tau_m}A\delta w \tag{60}$$

Similarly, the triangulated attribution provides an extra contribution that we denote as

$$\epsilon g_w \delta w, \quad \text{with} \quad g_w = \frac{\partial g}{\partial w}|_{w^*} \tag{61}$$

Considering that the direct sensitivity with respect to η is negligible (i.e. $\partial f/\partial\eta \approx 0$ because $\phi(0) = 0$, the linearized weight equation becomes

$$\frac{\mathrm{d}\delta w}{\mathrm{d}t} \approx [-\frac{\eta^*}{\tau_m}A + \epsilon g_w]\delta w \tag{62}$$

Thus, the effective linear coefficient in the w direction is $B = -\frac{\eta^*}{\tau_m}A + \epsilon g_w$. For the plasticity dynamics, linearizing the variable plasticity around w^* and η^* gives

$$\tau_d\frac{\mathrm{d}\delta\eta}{\mathrm{d}t} \approx -\delta\eta - \frac{1}{(\eta_{max} - w^*)^2}\delta w \tag{63}$$

Solving in a similar fashion to our previous derivation for the standard STDP system, the full two-dimensional linearized system in $(\delta w, \delta\eta)$ is then described by the Jacobian

$$\mathbf{J} = \begin{pmatrix} B & 0 \\ -\frac{1}{(\eta_{max}-w^*)^2\tau_d} & -\frac{1}{\tau_d} \end{pmatrix} \tag{64}$$

Again, referring to our previous derivation through Schur's identity, because the matrix is lower-triangular, its eigenvalues are provided by simply,

$$\lambda_1 = B = -\frac{\eta^*}{\tau_m}A + \epsilon g_w \quad \text{and} \quad \lambda_2 = -\frac{1}{\tau_d} \tag{65}$$

Since $\lambda_2 < 0$ for any positive τ_d, stability of the full system can be completely determined by the sign of λ_1. In the standard STDP case (i.e. $\epsilon = 0$), we had $-\frac{\eta^*}{\tau_m}A > 0$, implying instability. Thus, when the triangulated attribution mechanism contributes a sufficiently negative term, or that η_{max} is defined to influence A, we ensure local stability. This occcurs when

$$-\frac{\eta^*}{\tau_m}A + \epsilon g_w = -\frac{\eta^*}{\tau_m}\cdot\frac{\omega(-\frac{\theta}{wk\tau_m}}{\tau_m w^2 k\cdot(1+\omega(-\frac{\theta}{wk\tau_m}))} + \epsilon g_w < 0 \tag{66}$$

Geometrically, one may view the weight update as a vector field on the real line. In the standard model the field near w^* points outwards (unstable); the triangulated attribution acts as an additional "restorative" vector that effectively tilts the field inward, thereby reducing the net slope. We can invoke geometric arguments by comparing the phase portraits: if the combined directional derivative at w^* is negative, trajectories near w^* converge to it (i.e. the fixed point is an attractor). In this way, the triangulated mechanism "rotates" the flow in weight space so that the previously repulsive fixed point becomes locally attracting. Additionally, we can extend these findings to the multilayer case through use of the block Jacobian, as described in the first section, where the eigenvalues of synapses of layer l are determined by

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$$\lambda_w^l = -\frac{\eta^{*(l)}}{\tau_m} A^l + \epsilon g_w^l \quad \text{and} \quad \lambda_\eta^l = -\frac{1}{\tau_d} \tag{67}$$

3.3.3 Saddle Distributions and Free Probability

With this understanding of the network's eigenvalues, we can go on to characterize convergence and prove that MesoNet balances chaotic and stable phases to eventually reach an optimal solution. The critical aspect of our proposed algorithms are the saddle point distributions that emerge from the altered eigenvalues. Let x^* be an equilibrium point of a continuously differentiable autonomous dynamical system

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = f(\mathbf{x}), \quad \text{where } \mathbf{x} \in \mathbf{R} \tag{68}$$

such that $f(x^*) = 0$. Let $\mathbf{J} = Df(x^*)$ denote the Jacobian matrix of f evaluated at x^*, and let $\{\lambda_1, ..., \lambda_n\}$ be the eigenvalues of $\mathbf{J}$. Then, we declare x^* as a saddle point if the spectrum of $\mathbf{J}$ contains at least one eigenvalue with a positive real part and at least one eigenvalue with a negative real part, i.e.,

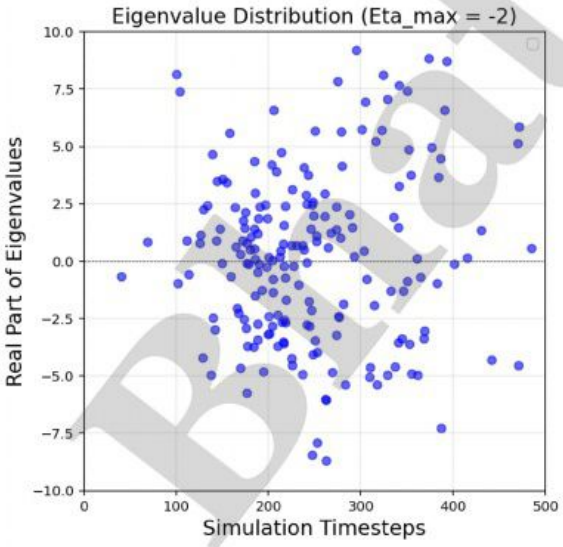


Figure 5: Randomly sampled synaptic eigenvalues from a three neuron (MesoNet-like) network over 500 timesteps, provided gaussian noise input.

$$\exists\, i, j \in \{1, n\} \text{ such that } \mathrm{Re}(\lambda_i) > 0 \text{ and } \mathrm{Re}(\lambda_j) < 0. \tag{69}$$

We can now determine the evolution of the locations of such saddle points. Reconsider the Lyapunov-like energy function defined in our previous analysis.

$$E(w, \eta) = \frac{1}{2}[\Delta t(w)]^2 + \frac{1}{2}(\eta - \eta^*)^2 \tag{70}$$

Note, again, that this function measures both the "timing error" (which STDP naturally pushes to zero), as well as the deviation of η from its steady state value η^*. Additionally it satisfies the requirement of positive definiteness and being decrescent. We differentiate E along its trajectories of the system as

$$\dot{E} = \frac{\partial E}{\partial w}\dot{w} + \frac{\partial E}{\partial \eta}\dot{\eta} = \Delta t(w)\frac{\mathrm{d}\Delta t}{\mathrm{d}w}(w)\dot{w} + (\eta - \eta^*)\dot{\eta} \tag{71}$$

We may now incorporate MesoNet's weight and plasticity dynamics. By proper design, Variable Plasticity and Triangulated Attribution are chosen such that they "counteract" positive expansion that occurs in the standard STDP rule. In particular, analysis (or numerical verification) shows that the corrective term $\epsilon g(w, \Delta t(w))$ is tuned such that the net derivative in the w-direction

$$\lambda_w(w, \eta) = \eta\phi'(\Delta t(w))\frac{\mathrm{d}\Delta t}{\mathrm{d}w}(w) + \epsilon g_w(w, \Delta t(w)), \tag{72}$$

tends to be negative as the network continues to evolve over time, scanning over a large domain of minima. Thus, we can conclude that over the region of interest, as the network becomes convergent (through the ϵg factor), energy is non-increasing. i.e.

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$$\dot{E}(w,\eta) \leq 0 \tag{73}$$

for all $(w,\eta) \neq (w^*,\eta^*)$, with equality holding only on the set

$$V = \{(w,\eta) \in \mathbf{R}^2 : \dot{E}(w,\eta) = 0\} \tag{74}$$

By LaSalle's Invariance Principle, every trajectory starting in a compact subset of the domain converges to the largest invariant subset of E. As the larges invariant set in E remains the single point (w^*,η^*), it follows that all trajectories converge asymptotically to (w^*,η^*), while minor invariant sets, such as saddles, shape trajectories. In a multilayer network, each synapse is governed locally by the two-dimensional dynamics we analyzed. The overall system can be described by a large block-structured Jacobian matrix $\mathbf{J}$ that collects contributions from all synapses across layers. For a given layer l, denote the linearized Jacobian (for each synapse) in the w direction as

$$J_l^w = \eta^{*l}\phi'(\Delta t(w^*))\frac{\mathrm{d}\Delta t}{\mathrm{d}t}(w^*) + \epsilon g_w^l. \tag{75}$$

To review, for each synapse in layer l, the two eigenvalues are

$$\lambda_w^l = d^l = -\frac{\eta^*}{\tau_m}A^l + \epsilon g_w^l \;\text{ and }\; \lambda_\eta^l = -\frac{1}{\tau_d} \tag{76}$$

In realistic networks the paramaters (e.g. the local slope A^l or the correction g_w^l) exhibit heterogeneity and there are additional fluctuations due to network interactions. We now decompose each layer's contribution to the Jacobian into a determinisitc part and a fluctuation part. For instance, for the weight-direction we write

$$\Lambda^l = d^l I_{N_l} + X^l \tag{77}$$

where

- $d^l = -\frac{\eta^*}{\tau_m}A^l + \epsilon g_w^l$ is the deterministic shift for layer l mean behavior (i.e. the average expansion rate in the w-direction)
- I_{N_l} is the identity matrix of size $N_l \times N_l$
- X^l is a random fluctuation matrix that accounts for inhomogeneities and coupling effects. In our case, X^l is drawn from a Wigner ensemble so that its spectral density obeys the semicircular law with variance σ_l^2.

Because the overall Jacobian is block-diagonal (for the eigenvalues) in each group, the spectrum in each layer is given by the free additive convolution of the delta-distribution at d^l with the semicircular law of X^l. That is, the eigenvalue density for the w direction in layer l is

$$\rho_{\Lambda^l} = \frac{1}{2\pi\sigma_l^2}\sqrt{4\sigma_l^2 - (\lambda - d^l)^2} \;\; for \;\; \lambda \in [d^l - 2\sigma_l, d^l + 2\sigma_l] \tag{78}$$

and the contribution from the variable plasticity is simply a point mass at

$$\lambda_\eta = -\frac{1}{\tau_d} \tag{79}$$

with multiplicity equal to the number of synapses. Finally, the full spectrum of the Jacobian $\mathbf{J}$ is given by the union (weighted by the number of synapses in each layer) of the spectra from the two groups:

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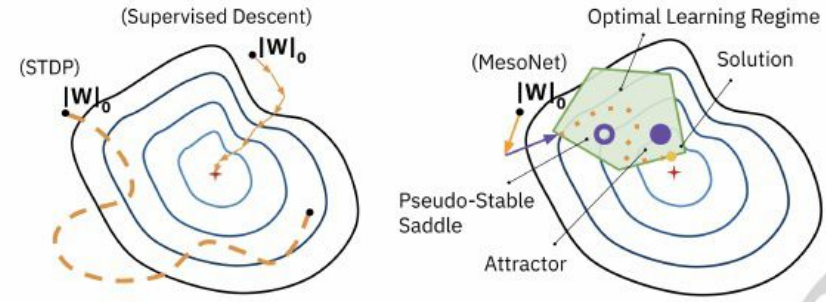


Figure 6: A visualization of MesoNet's unsupervised learning algorithm within weight space.

$$\rho_J(\lambda) = \frac{N_1}{N_1 + N_2}\rho_{\Lambda^1}(\lambda) + \frac{N_2}{N_1 + N_2}\rho_{\Lambda^2}(\lambda) \tag{80}$$

augmented by the point mass at $\lambda = -1/\tau_d$. Consequently, we can describe the saddle distribution through analyzing the eigenvalues of the system. When the deterministic shift is negative enough compared to the width of the fluctuations, then almost surely all eigenvalues lie in the negative half-plane, guaranteeing exponential convergence of perturbations and validating the MesoNet algorithms in a large time-scale setting, while remaining within the optimal regime. This concludes the MesoNet Central Conjecture.

4 Experiments

4.1 Implementation

In order to maximize improvement of feature extraction and accuracy, MesoNet is architecturally designed to leverage the characteristics of Variable Plasticity and Triangulated Attribution. To accomplish this, we utilize the foundations of the model proposed by Meng et al. (2021) [14]t o create a Split and Merge Architecture derived from the highly popularized Inception model from the ANN setting [22]. In this framework, input images split into multiple pathways to be processed by various convolutional filters (eg. 3×3, 5×5, 7×7) before being concatenated back together. This methodology has two major benefits: (1) The split-and-merge module can process multi-scale spatial information to improve feature detection. (2) It improves the network's parallelism as various pathways can operate concurrently. MesoNet was designed with the Brian2 simulator; preliminary designs are available here: **https://github.com/blizzard-labs/NMCtests**

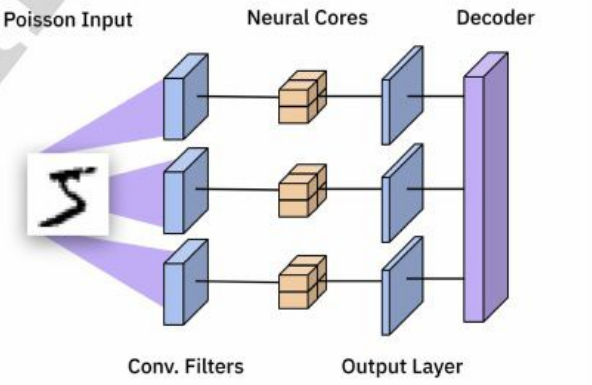


Figure 7: SNNs process information asynchronously through sparse biological spikes with spike timings encoding information.

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4.2 Experimental Design

In this section, we conduct various experiments on benchmark datasets to evaluate the functional performance of MesoNet. These procedures were formulated to answer the following question: *How does the performance of MesoNet compare with state-of-the-art SNNs and ANNs in unsupervised image classification tasks?*

The following models and datasets were utilized for evaluation:

- *Models:* MesoNet, Diehl & Cook's 2015 FC Network, Convolutional-SNN, Adaptive Synaptic Plasticity Model, Inviariant Information Clustering (Non-SNN network)
- *Datasets:* MNIST, CIFAR-10, CIFAR-100, F-MNIST, E-MNIST

For direct comparison, we used the basic 2 layer connection proposed by (Diehl & Cook, 2015) as a baseline and will be comparing MesoNet to the Adaptive Synaptic Plasticity model [16] and CSNN [11]. We will also be comparing MesoNet to the Invariant Information Clustering (IIC) algorithm, the SOTA unsupervised ANN at the time of writing proposed by Ji et al. (2019). [9] Additionally, we held the following hyperparameters constant for all models across the datasets, as described by the following table.

Parameter	MNIST	CIFAR10	CIFAR100	F-MNIST	E-MNIST
Encoding Length	300	350	450	300	300
Learning Rate	0.01	0.005	0.001	0.01	0.01
Membranc Time Constant	2 ms	2 ms	2 ms	2 ms	2 ms
Synapse Time Constant	1 ms	1 ms	1 ms	1 ms	1 ms

Figure 8: Standardized testing parameters for evaluation.

In our testing, we used a standard procedure similar to the one used by Diehl & Cook (2015) [3]: A static image is presented for training to the SNN for 350ms before a 150ms phase which allows neurons to settle down to their original state. The networks were trained with a single pass through a training segment of each dataset (70%) such that there was in total one image per iteration (eg. 50,000 iterations for MNIST). After training, weights were frozen and accuracy was calculated based upon the remaining images (30%) in each dataset.

Accuracy was measured across epochs with a running loss function and robustness as the maximum number of synapses or neurons removed before observing a drop in accuracy of 5%. Here we report the results of the evaluation:

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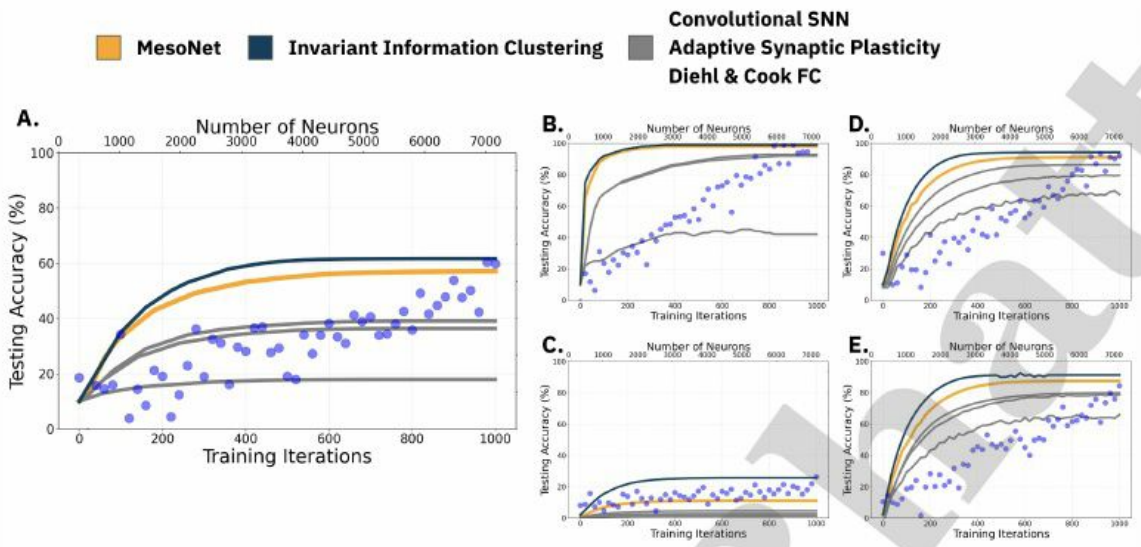


Figure 9: Classification accuracies (for all models) and neuronal robustness (for MesoNet)- A: CIFAR-10, B: MNIST, C: CIFAR-100, D: F-MNIST, E: E-MNIST

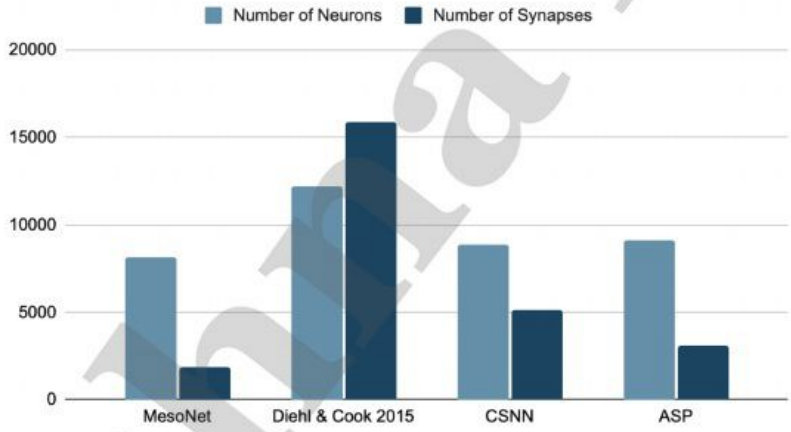


Figure 10: Synaptic/neuronal robustness on MNIST

5 Discussion

5.1 Summary of Contributions

This work addresses fundamental challenges in spiking neural network learning through a principled, theory-driven approach. By applying dynamical systems theory to analyze STDP-based learning, we uncovered that standard unsupervised SNNs operate in an inherently unstable regime where the learning rate η acts as a bifurcation parameter controlling system convergence. Our theoretical framework revealed that conventional STDP implementations place networks in a dissipative regime where perturbations amplify exponentially, explaining the longstanding scalability and generalization issues that have plagued unsupervised SNNs.

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Building on these insights, we introduced two complementary mechanisms—Variable Plasticity and Triangulated Attribution—that dynamically regulate network dynamics to maintain operation near the edge of chaos. These algorithms work synergistically: Variable Plasticity adapts learning rates based on local weight dynamics to prevent runaway instability, while Triangulated Attribution enables non-local credit assignment by propagating updates to upstream synapses when similar timing errors are detected across multiple connections. Together, these mechanisms form and annihilate saddle points in the weight-plasticity space, guiding the network through optimal learning trajectories.

The resulting architecture, MesoNet, demonstrates that these theoretical principles translate to substantial practical improvements. Our experiments show that MesoNet achieves 56.6% accuracy on CIFAR-10, representing an 18.57 percentage point improvement over previous state-of-the-art unsupervised SNNs, while operating within 5.05 percentage points of contemporary unsupervised ANNs. Critically, MesoNet accomplishes this while maintaining robustness, demonstrating that theoretical rigor and practical efficiency can be simultaneously achieved.

5.2 Interpretation of Results

5.2.1 Performance Across Datasets

The experimental results reveal distinct patterns in how MesoNet handles datasets of varying complexity. On MNIST, where spatial features are relatively simple and well-defined, MesoNet achieves near-perfect performance, suggesting that our algorithms successfully capture and maintain representations of basic visual patterns. The substantial performance gap between MesoNet and baseline STDP approaches on this dataset validates our theoretical prediction that maintaining operation near the edge of chaos enables more stable feature learning.

The CIFAR-10 results are particularly noteworthy. The 56.6% accuracy represents a fundamental advancement in unsupervised SNN capability, demonstrating for the first time that STDP-based networks can learn meaningful hierarchical representations from complex natural images. The 18.57 percentage point improvement over previous methods suggests that our theoretical framework has identified and addressed a core limitation of conventional approaches. However, the remaining gap to state-of-the-art ANNs (approximately 5 percentage points) indicates that further refinements are needed to fully bridge the performance divide.

On CIFAR-100, F-MNIST, and E-MNIST, we observe consistent improvements over baseline methods, though the absolute accuracies vary with task complexity. The hierarchical structure of CIFAR-100 (100 fine-grained categories) presents a particularly challenging test case for unsupervised learning, and while MesoNet shows improvement, the results suggest that additional architectural innovations may be required to handle extreme fine-grained categorization tasks.

The robustness experiments reveal an interesting property of MesoNet's learned representations. The network maintains performance under significant neuronal and synaptic removal, suggesting that Variable Plasticity and Triangulated Attribution promote distributed representations rather than brittle, localized features. This robustness likely emerges from the dynamic saddle distributions: by continuously forming and annihilating saddles, the network explores multiple representational strategies and converges to solutions that leverage redundancy across the architecture.

5.2.2 Dynamical Systems Insights

Our Lyapunov exponent analysis provides a compelling explanation for MesoNet's success. By operating near the edge of chaos—where the dominant Lyapunov exponent approaches zero—the network balances the competing demands of stability and adaptability. In chaotic regimes, small

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differences in input amplify exponentially, preventing coherent learning; in overly stable regimes, the network becomes trapped in local minima and fails to explore the solution space effectively. The edge of chaos represents a "sweet spot" where the network can both retain learned features (stability) and incorporate new information (adaptability).

The saddle point dynamics revealed by our free probability analysis offer a geometric interpretation of this balance. Rather than smoothly descending toward a single global minimum, MesoNet's learning trajectory navigates a landscape populated by transient saddle points. These saddles act as decision points where the network can transition between different representational strategies, enabling escape from poor local minima while gradually converging toward high-quality solutions.

5.3 Limitations and Future Work

5.3.1 Theoretical Extensions

While our dynamical systems framework provides substantial insight, several theoretical questions remain open. First, our stability analysis relies on linearization near fixed points, which provides local guarantees but does not fully characterize global convergence properties. Future work should explore whether global stability results can be obtained, perhaps using Lyapunov function techniques extended to the full nonlinear system. Second, our free probability analysis assumes that fluctuations in the Jacobian follow a Wigner ensemble with semicircular spectral density. While this assumption is reasonable for large, randomly initialized networks, real learning dynamics may induce correlations that violate this independence assumption. Investigating the spectral properties of Jacobians during actual learning could reveal whether more sophisticated random matrix models (e.g., heavy-tailed or correlated ensembles) are needed. Third, the quadratic spike approximation, while tractable and effective in our experiments, introduces approximation error that accumulates over deep layers. Developing higher-order approximations or alternative analytical techniques that avoid the Lambert W function's multi-valuedness could improve accuracy for very deep architectures.

5.3.2 Architectural Innovations

The current MesoNet implementation uses a split-and-merge architecture inspired by the Inception model. While this design effectively leverages multi-scale processing, several architectural extensions warrant exploration:

- **Deeper Hierarchies:** Our experiments focused on relatively shallow networks. Investigating how Variable Plasticity and Triangulated Attribution scale to networks with 10+ layers would test whether the theoretical guarantees of stability extend to truly deep architectures. Preliminary analysis suggests that cascading effects in deep networks may require layer-specific tuning of hyperparameters like η_{min} and ϵ.
- **Recurrent Connections:** The current feedforward architecture does not exploit the temporal processing capabilities inherent to SNNs. Incorporating recurrent connections could enable the network to learn temporal dependencies and sequential patterns. However, recurrence introduces additional dynamical complexity that may require extensions to our stability analysis.
- **Attention Mechanisms:** Recent advances in ANNs have demonstrated the power of attention mechanisms for selectively focusing on relevant features. Developing spike-based attention mechanisms compatible with STDP learning could further enhance MesoNet's representational capacity, particularly for tasks requiring long-range dependencies.

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5.3.3 Learning Algorithm Refinements

Several refinements to Variable Plasticity and Triangulated Attribution could improve performance:

- **Adaptive Time Constants:** The current implementation uses fixed time constants τ_d for plasticity decay. Allowing these constants to adapt based on learning progress (e.g., increasing τ_d as features stabilize) could improve convergence speed and final accuracy.
- **Hierarchical Attribution:** Triangulated Attribution currently operates within individual layers. Extending this mechanism to propagate updates across multiple layers simultaneously could enable more effective end-to-end feature learning. However, this would require careful analysis to ensure stability is maintained.
- **Meta-Learning Hyperparameters:** The hyperparameters η_{min}, η_{max}, ϵ, and κ significantly impact performance, and their optimal values likely vary across datasets and architectures. Developing meta-learning approaches to automatically tune these parameters could make MesoNet more broadly applicable without extensive manual tuning.

5.3.4 Biological Plausibility

While STDP is biologically inspired, Variable Plasticity and Triangulated Attribution introduce mechanisms whose biological substrates remain unclear. Variable Plasticity could potentially correspond to metaplasticity phenomena observed in real neurons, where past activity modulates future plasticity. Triangulated Attribution's non-local updates are more challenging to map onto known biological mechanisms, though processes like retrograde signaling and dendritic computation may provide partial analogies. Future work should investigate whether these algorithms can be reformulated in more strictly biological terms. If so, MesoNet could serve as a computational model for understanding how biological neural networks achieve stable, efficient learning. Conversely, if these mechanisms cannot be grounded in biology, this would suggest interesting divergences between optimal computational strategies and the constraints under which biological brains evolved.

5.4 Broader Implications

5.4.1 Bridging Theory and Practice in Neuromorphic Computing

The neuromorphic computing community has long recognized a tension between biological inspiration and computational performance. Early SNNs closely mimicked biological details but struggled with practical tasks; more recent approaches have improved performance by incorporating techniques from deep learning, often at the cost of biological plausibility. MesoNet suggests a middle path: by developing theory that explains why biological mechanisms succeed (or fail), we can design algorithms that are simultaneously effective and theoretically principled.

This theory-driven approach has implications beyond SNNs. Many machine learning advances have been empirically driven, with theoretical understanding lagging behind practical success. Our work demonstrates that investing in theoretical foundations—even for complex, nonlinear systems like SNNs—can yield novel algorithms that outperform purely empirical approaches.

5.4.2 Understanding Learning in Artificial and Biological Systems

Finally, our dynamical systems analysis offers insights that may extend beyond the specific algorithms we developed. The observation that learning systems can be characterized by their position in a

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spectrum from chaotic to stable regimes has parallels in neuroscience, where concepts like "criticality" and "balanced excitation-inhibition" describe similar phenomena in biological brains.

The saddle point dynamics we identified suggest that effective learning may inherently involve navigating complex, non-convex optimization landscapes rather than smoothly descending gradients. This perspective aligns with recent theoretical work on the loss landscapes of deep neural networks and may point toward fundamental principles governing learning in both artificial and biological systems.

5.5 Conclusion

This work establishes that principled theoretical analysis can unlock improvements in spiking neural network performance. By identifying the edge of chaos as an optimal learning regime and developing algorithms that maintain this operating point, we have advanced the state-of-the-art in unsupervised SNN learning while preserving the energy efficiency that makes neuromorphic computing attractive. MesoNet is another step toward realizing the promise of neuromorphic computing: intelligence that approaches biological efficiency. While challenges remain—particularly in scaling to very deep networks and closing the remaining performance gap with supervised ANNs—our results demonstrate that unsupervised SNNs can handle complex real-world tasks. As neuromorphic hardware continues to mature and the demand for energy-efficient AI intensifies, approaches like MesoNet may transition from research curiosities to practical solutions for edge intelligence, medical devices, space exploration, and beyond. The path forward requires continued dialogue between theory and experiment. Our dynamical systems framework suggests further algorithmic innovations, and experimental validation of these ideas will, in turn, refine the theory. By maintaining this iterative cycle, the neuromorphic computing community can build toward systems that combine the computational power of modern AI with the efficiency and robustness of biological brains.

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Supplementary Materials | MesoNet: Optimizing Spiking Neural Networks with Dynamic Saddle Distributions

1 Quadratic Spike Approximation

In order to overcome the issues associated with utilizing the Lambert W function, I propose a quadratic spike approximation which provides a deterministic method of computing spike timing differences. These results can be extended to network implementations as well as dynamical systems analysis. Beginning with Eq. (7), we can take the first derivate through simple calculation as

$$f'(x) = wx \ \cdot \ -\frac{1}{\tau_m} e^{-\frac{x}{\tau_m}} + we^{-x} \tag{1}$$

Setting this to 0, we find that $f(x)$ obtains a relative maximum at $x = \tau_m$. Thus this marks the bounds for possible spiking as $[0, \tau_m]$. As such, we need to approximate the spiking only up to τ_m. Expanding Eq. (8) into a MacLaurin series about $x = 0$, we find that

$$e^{-x/\tau_m} = \sum_{n=0}^{\infty} \frac{(-x/\tau_m)^n}{n!} \ \text{ and } \ T_N(x) = \sum_{n=0}^{N} \frac{(-x/\tau_m)^n}{n!} \tag{2}$$

Thus, the truncated approximation for spike timing is given by

$$f(x) \approx w \cdot k \cdot x \cdot T_N(x) = \theta \tag{3}$$

We must now determing the minimum order N such that our quadratic approximation provides accuracy up to the operating time interval. For the purposes of this project, that will be set to 0.5. Taking the Lagrange error,

$$R_{N+1} = \frac{e^{-c/\tau_m}}{(n+1)!}(-\frac{x}{\tau_m})^{n+1}, \ \text{ with } \ c \in [0, x] \tag{4}$$

Thus at $x = \tau_m$, we have

$$|R_{N+1}(\tau_m)| \leq \frac{e^{-c/\tau_m}}{(n+1)!}|\frac{\tau_m}{\tau_m}|^{n+1} \leq \frac{1}{(n+1)!} \tag{5}$$

Since c is bounded by $[0, \tau_m]$, we can declare $e^{-c/\tau_m} \leq 1$. This gives

$$\Delta f(x) \approx wkx \cdot R_{N+1}(x) \tag{6}$$

where

$$|\Delta f(x)| \leq wk\tau_m \cdot \frac{1}{(n+1)!} \tag{7}$$

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Going back to Eq. (12), notice again that at $x = \tau_m$ we have

$$f'(\tau_m) = wke^{-1}(1-1) = 0 \tag{8}$$

Because the first derivative vanishes at $x = \tau_m$, the usual linear error propogation formula (utilizing $1/|f'(x)|$) does not apply. Instead, we use a second-order (quadratic) error propogation. For a function with a simple zero where the first derivative vanishes but the second derivative is nonzero, a small change Δf in the function produces a shift in the root of

$$\Delta x \approx \sqrt{\frac{2|\Delta f|}{|f''(\tau_m)|}} \tag{9}$$

Computing f'' with product rule, a short calculation reveals

$$\begin{aligned} f'(x) &= wke^{-x/\tau_m}(1 - \frac{x}{\tau_m}) \\ f''(x) &= -wke^{-x/\tau_m}(\frac{2 - x/\tau_m}{\tau_m}) \end{aligned} \tag{10}$$

At $x = \tau_m$,

$$f''(\tau_m) = -wke^{-1}(\frac{2-1}{\tau_m}) = -\frac{wk}{e\tau_m} \tag{11}$$

Thus, in absolute value, the error in x is nearly

$$|\Delta x| \approx \sqrt{\frac{2|\Delta f(\tau_m)|}{|f''(\tau_m)|}} \leq \sqrt{\frac{2[wk\tau_m \cdot 1/(n+1)!]}{wk(e\tau_m)^{-1}}} = \sqrt{\frac{2\tau_m^2 e}{(n+1)!}} \tag{12}$$

To have computed x within 0.5 of τ_m, we require

$$\sqrt{\frac{2e\tau_m^2}{(n+1)!}} \leq 0.5 \tag{13}$$

Rearranging the expression gives

$$\frac{2e\tau_m^2}{(n+1)!} \leq 0.25 \quad \Rightarrow \quad (n+1)! \geq 8e\tau_m^2 \tag{14}$$

where n now provides the minimum-ordered approximation required to approximate spike timing within the operating interval. This concludes the proof for quadratic spike approximation.

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2 Linear Stability Analysis

We can now extend these findings to develop a complete model for the behavior of the 1:1 connection. Consider the STDP equation from Eq. (5). This can be rearranged and simplified to the basic form

$$\Delta w(\Delta t(w)) = \text{sgn}(\Delta t(w)) \cdot \eta \cdot \exp(-\text{sgn}(\Delta t(w)) \cdot \frac{\Delta t(w)}{\tau_m}) \tag{15}$$

where sgn is the "sign" function such that $\text{sgn}(x) = |x|/x$. We can also recast STDP dynamics by defining the Poisson-based spike time difference

$$\Delta t(w) = n(\frac{1}{k_{post}(w)} - \frac{1}{k}) \tag{16}$$

so that the sign of Δt now directly reflects whether the post-synaptic interspike interval is longer or shorter than the pre-synaptic one. (Here n indexes the spike pair). For a given connection, the fixed point is reached when $\Delta t(w) = 0$. This is because the magnitude of update is nonzero unless the exponential factor acts on a zero argument, the fixed point of the dynamics occurs when

$$\Delta t(w) = 0 \tag{17}$$

That is, by employing Eq. (9), when

$$\frac{1}{k_{post}(w} = \frac{1}{k} \Rightarrow k_{post}(w) = k = -\tau_m \omega(-\frac{\theta}{wk\tau_m}) \tag{18}$$

Considering that $W(z)e^{W(z)} = z$ as per the Lambert W Function, we substitute

$$z = -\frac{\theta}{w^* k\tau_m} \Rightarrow -\frac{k}{\tau_m}\exp(-\frac{k}{\tau_m}) = -\frac{\theta}{w^* k\tau_m} \tag{19}$$

Thus, the fixed point weight is provided by

$$w^*(k)^2\exp(\frac{-k}{\tau_m}) \rightarrow w^* = \frac{\theta e^{k/\tau_m}}{k^2} \tag{20}$$

Now, we go on to analyze the stability of the fixed point. Firstly, we can recast the weight dynamics as a differential equation

$$\frac{\mathrm{d}w}{\mathrm{d}t} = f(w) \tag{21}$$

where

$$f(w) = \text{sgn}(\Delta t(w)) \cdot \eta \cdot \exp(-\frac{|\Delta t(w)|}{\tau_m}) \tag{22}$$

Because the only zero of $f(w)$ is at $w = w^*$ (since the exponential term is always positive), we linearize f around w^*. Letting

$$w = w^* + \delta, \tag{23}$$

we have

$$f(w) \approx f(w^*) + f'(w)\delta \tag{24}$$

Since $f(w^*) = 0$, the linearized dynamics are governed by

$$\frac{\mathrm{d}\delta}{\mathrm{d}t} = f'(w^*)\delta \tag{25}$$

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Thus, the fixed point is stable if $f'(w^*) < 0$ and unstable if $f'(w^*) > 0$. Further, we can determine $f'(w^*)$ explicitly. Consider $\Delta w(\Delta t(w))$ from Eq. (26). Define

$$g(x) = \text{sgn}(x)\eta \text{exp}(-\frac{\text{sgn}(x)x}{\tau_m}) \tag{26}$$

Because at the fixed point $\Delta t(w^*) = 0$, we can compute

$$\Delta g'(0) = \lim_{x \to 0} \frac{g(x) - g(0)}{x} \tag{27}$$

From further evaluation, approaching from above and below,

$$g'(0) = -\frac{\eta}{\tau_m} \tag{28}$$

Then, by chain rule,

$$f'(w^*) = g'(0)\frac{\mathrm{d}\Delta t}{\mathrm{d}w}|_{w^*} = -\frac{\eta}{\tau_m}\frac{\mathrm{d}\Delta t}{\mathrm{d}w}|_{w^*} \tag{29}$$

Recall Eq. (29), and since k can be assumed as constant for any given firing cycle,

$$\frac{\mathrm{d}\Delta t}{\mathrm{d}w} = -\frac{n}{(k_{post}(w))^2}\frac{\mathrm{d}k_{post}}{\mathrm{d}w} \tag{30}$$

Differentiation k_{post} with respect to w (using chain rule and the derivative of the Lambert W function, $W'(z) = \frac{W(z)}{z(1+W(z))}$), we can obtain

$$\frac{\mathrm{d}k_{post}}{\mathrm{d}w} = \frac{\tau_m \omega(u)}{w(1+\omega(u))} \text{ with } u = \frac{-\theta}{wk\tau_m} \tag{31}$$

At the fixed point, we have $k_{post}(w^*) = k$, so

$$-\tau_m \omega(u^*) = k \Rightarrow \omega(u^*) = -\frac{k}{\tau_m} \tag{32}$$

Thus, at w^*,

$$\frac{\mathrm{d}k_{post}}{\mathrm{d}w}|_{w^*} = \frac{\tau_m(-\frac{k}{\tau_m})}{w^*(1-\frac{k}{\tau_m})} \tag{33}$$

Substituting back into the derivative of $\Delta t(w)$

$$\frac{\mathrm{d}\Delta t}{\mathrm{d}w}|_{w^*} = -\frac{n}{k^2}[-\frac{k}{w^*(1-\frac{k}{\tau_m})}] = \frac{n}{w^*k(1-\frac{k}{\tau_m})} \tag{34}$$

Thus, the final linearization is given by

$$f'(w^*) = -\frac{\eta}{\tau_m}\frac{n}{w^*k \cdot (1-\frac{k}{\tau_m})} \tag{35}$$

The evolution of f(w) depends entirely on the factor $1 - \frac{k}{\tau_m}$ and the learning rate parameter η. In many biophysical contexts, the membrane time constant τ_m is small compared to the effective time scale set by k_{pre}, so that typically,

$$\frac{k}{\tau_m} > 1 \Rightarrow 1 - \frac{k}{\tau_m} < 0 \tag{36}$$

This makes $f'(w^*) > 0$, so that in the standard STDP-LIF SNN 1:1 connection, small perturbations in weight grow exponentially and the fixed point w^* is entirely unstable.

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3 Compiled Data Table

MESONET (SPLIT AND MERGE)					
Dataset	**Accuracy (%)**	**STD dev.**	**F1-Score**	**Min. Synapses (Robustness)**	**Energy Consumption (kWh)**
MNIST	98.11	0.17	0.981	1900	1.865
E-MNIST	87.51	0.21	0.873	-	-
F-MNIST	91.23	0.23	0.91	-	-
CIFAR-10	56.6	0.49	0.559	-	-
CIFAR-100	11.16	0.56	0.107	-	-
CONVOLUTIONAL SNN (SLAYER)					
Dataset	**Accuracy**	**STD dev.**	**F1-Score**	**Min. Synapses (Robustness)**	**Energy Consumption (kWh)**
MNIST	92.53	0.21	0.924	5100	3.612
E-MNIST	64.98	0.23	0.645	-	-
F-MNIST	86.33	0.22	0.861	-	-
CIFAR-10	38.03	0.35	0.375	-	-
CIFAR-100	4.54	0.76	0.042	-	-
ADAPTIVE SYNAPTIC PLASTICITY					
Dataset	**Accuracy**	**STD dev.**	**F1-Score**	**Min. Synapses (Robustness)**	**Energy Consumption (kWh)**
MNIST	93.66	0.23	0.935	3100	1.921
E-MNIST	85.73	0.36	0.854	-	-
F-MNIST	79.21	0.42	0.787	-	-
CIFAR-10	36.41	0.95	0.359	-	-
CIFAR-100	2.63	1.32	0.024	-	-
SIMPLE 2-LAYER FC					
Dataset	**Accuracy**	**STD dev.**	**F1-Score**	**Min. Synapses (Robustness)**	**Energy Consumption (kWh)**
MNIST	42.87	0.54	0.421	15900	1.322
E-MNIST	78.17	0.46	0.775	-	-
F-MNIST	68.29	0.49	0.678	-	-
CIFAR-10	17.92	1.27	0.173	-	-
CIFAR-100	1.2	1.64	0.01	-	-
INVARIANT INFORMATION CLUSTERING					
Dataset	**Accuracy**	**STD dev.**	**F1-Score**	**Min. Synapses (Robustness)**	**Energy Consumption (kWh)**
MNIST	99.71	0.08	0.997	N/A	3.984
E-MNIST	91.29	0.22	0.911	-	-
F-MNIST	94.29	0.17	0.942	-	-
CIFAR-10	61.7	0.47	0.614	-	-
CIFAR-100	25.7	0.53	0.252	-	-

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